

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering on Imbalanced Mixtures

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Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. We study unlabeled selection of synthetic samples by a denoising likelihood proxy in self-consuming DDPM training on imbalanced mixtures. Define the minority-mass ratio $r = m^{(1)}/m^{(0)}$. Under matched real-mix ratio and matched synthetic budget, on a primary matrix of **eight** seed–imbalance cells,

$$\text{bottom-}k > \text{random-}k > \text{top-}k$$

holds on mean r (0.57/0.36/0.17), but a per- ρ split shows the full order is driven by $\rho=5$ (5/5 seeds) and does *not* hold seed-wise at $\rho=10$ (0/3). Paired contrasts vs. random- k ($n=8$) show large effects (Cohen’s $d_z \approx 1.0$; Wilcoxon $p \approx 0.042$; Holm-adjusted $p \approx 0.085$). A hypothesized “false stability” pattern—top- k *improving* sliced W_2 while minorities die—is **not observed** at generation 1 ($\Delta W_2 > 0$ on 8/8 cells; Wilcoxon $p \approx 0.014$). Directional transfer checks on 8D GMM and Digits-PCA (3 seeds) match the order. Code: <https://github.com/muhammadalamezb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse*: progressive loss of diversity and forgetting of rare modes [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024], injecting fresh real samples, and *filtering* synthetic data [Feng et al., 2024, Cai et al., 2025].

When the only available score is the generator’s own likelihood (or denoising loss), how should one select synthetics? Feng et al. [2024] show that *task verifiers* for labeled synthetic answers can prevent collapse. Cai et al. [2025] filter “unrealistic” diffusion samples via latent probes and report aggregate FID/precision/recall. Alemohammad et al. [2023] observe that sampling bias toward high-quality generations drives a quality–diversity tradeoff. What is missing is a **matched-budget, mode-exact** causal contrast among likelihood ranks.

What is new. Prior work describes quality–diversity tradeoffs and filtering heuristics. We isolate an *interventional* selection policy π (top- k vs. random- k vs. bottom- k) at matched synthetic count and matched real-mix α , and measure mode masses against known centers. The contribution is this controlled contrast and the resulting empirical minority-retention order—not a new named training algorithm or ImageNet-scale SOTA method.

Contributions.

1. A controlled self-consuming DDPM protocol on imbalanced GMMs with exact mode masses.
2. Evidence for a likelihood-rank ordering of minority retention under matched k and α , with an explicit per- ρ split.
3. Evidence that top- k does *not* improve sliced W_2 vs. random- k while minorities decline.
4. Directional transfer checks: 8D GMM and Digits-PCA with nearest class-mean occupancy.

2 Related work

Model collapse. Shumailov et al. [2023] document recursive training collapse and tail forgetting. Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Alemohammad et al. [2023] (*MAD*) highlight quality-biased sampling.

Verification and filtering. Feng et al. [2024] argue verification of synthesized *labels* prevents collapse. Cai et al. [2025] (*LSF*) filter synthetic images using latent-space realism scores. Neither isolates unlabeled ELBO top- k vs. random- k with mode-grounded metrics on continuous mixtures.

Temperature / diversity knobs. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we instead intervene on *training-set selection* inside self-consuming loops.

3 Method

3.1 Self-consuming loop

Let $f^{(g)}$ be a DDPM trained on $\mathcal{D}^{(g)}$. Sample synthetic set $\widehat{\mathcal{D}}^{(g+1)}$. Build the next train set by mixing a fixed real buffer $\mathcal{D}_{\text{real}}$ (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$ is mean denoising MSE over a t -grid (lower = higher likelihood). At matched synthetic count n_{syn} and real fraction α :

- **top- k / bottom- k / random- k :** select a pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores, then draw n_{syn} from that pool and mix with α real.
- **mix:** same α real mix, but the synthetic portion is a uniform subsample of *all* N synthetics (no rank filter; no k -pool stage).
- **replace:** training set is pure synthetic (collapse reference).

Thus **mix** $\neq$ **random- k** : random- k still uses a size- k subsample stage (matched to top/bottom); mix skips ranking entirely.

Table 1: Primary analysis cells (exactly eight seed–imbalance pairs). Seeds 3–4 were run only at $\rho=5$ (replication); $\rho=10$ uses seeds $\{0, 1, 2\}$ only.

$\rho \backslash$ seed	0	1	2	3	4
5	✓	✓	✓	✓	✓
10	✓	✓	✓	—	—

Table 2: Minority-mass ratio $r = m^{(1)}/m^{(0)}$ pooled over the eight-cell primary matrix (Table 1).

Policy	Mean r	Std	n
bottom_k	0.572	0.289	8
rand_k	0.360	0.173	8
mix	0.328	0.100	8
top_k	0.172	0.078	8

3.3 Metrics

Nearest-mean mode assignment on known centers (GMM means, or Digits empirical class means in PCA space). Report minority-mean mass m (modes $1 \dots K-1$), majority mass (mode 0), and

$$r = \frac{m^{(1)}}{m^{(0)}},$$

the minority-mass *ratio* (called retention even when $r > 1$). Sliced W_2 : average of squared 1D Wasserstein distances over **24** random projections; each comparison uses up to **400** synthetic and **400** held-out real points.

4 Experiments

4.1 Primary design (defines $n=8$)

$K=4$ modes, tiny MLP denoiser, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl` ($n=228$ records).

4.2 Primary retention results

We report the ratio r from §3.3. Table 2 averages r over all eight cells; Table 3 splits by ρ .

Interpretation. Pooled means hide a sharp split: at $\rho=5$ the ordering is unanimous (5/5). At $\rho=10$, minorities are already near floor at $g=0$, and the full order holds on **zero** of three seeds (top<rand on 2/3; bottom>rand on 1/3 only). Thus the pooled “7/8 top worse; 6/8 bottom better” counts are largely driven by the $\rho=5$ subgroup; we do **not** claim robust seed-wise ordering at $\rho=10$.

Paired retention contrasts vs. random- k on the pooled $n=8$ cells: top–rand mean $\Delta=-0.188$ (bootstrap 95% CI $[-0.30, -0.06]$; Cohen’s $d_z=-1.04$; Wilcoxon $p \approx 0.042$); bottom–rand mean $\Delta=+0.212$ (CI $[0.08, 0.35]$; $d_z=+1.00$; $p \approx 0.042$). Holm-adjusted $p \approx 0.085$ for both. We emphasize effect sizes/CIs and the $\rho=5$ directional evidence over uncorrected p -values.

Table 3: Per- ρ breakdown of mean r and seed-wise directional counts (generation 1). The full order holds at $\rho=5$ but not at $\rho=10$.

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	5	0.737	0.456	0.183	5/5	5/5 / 5/5
10	3	0.296	0.199	0.154	0/3	2/3 / 1/3

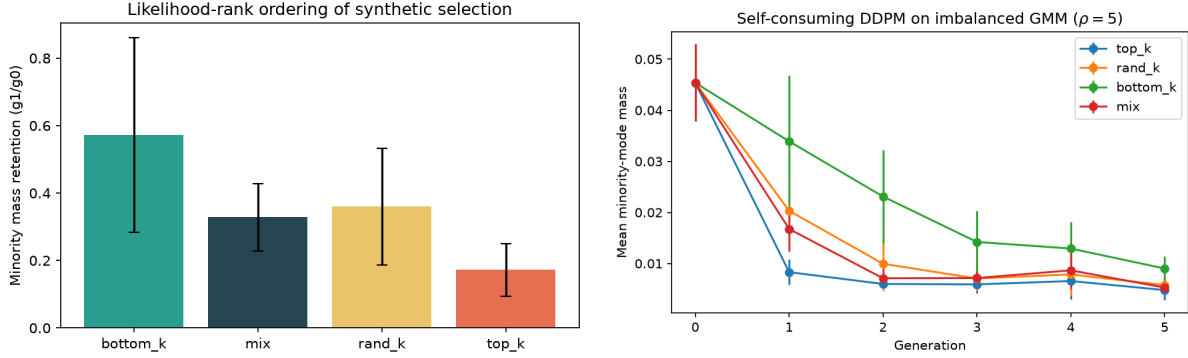


Figure 1: Left: retention by policy on the primary matrix. Right: minority mass vs. generation at $\rho=5$ (the regime where the order is stable).

4.3 Sliced W_2 (“false stability” check)

If top- k were “falsely stable,” we would expect better (lower) sliced W_2 than random- k while minorities die. Table 4 reports $\Delta W_2(\text{top} - \text{rand})$ by generation on the eight-cell matrix.

We report this as **evidence against** a top- k W_2 improvement under minority loss—not as a formal Neyman–Pearson falsification.

4.4 Real-mix α ablation (illustrative; $n=2$ seeds)

Separate exploratory ablation (`w2_fs_alpha.jsonl`; seeds $\{0, 1\}$, $\rho=5$, $G \leq 3$). With only two seeds, order-hold counts are preliminary and should not be treated as establishing a hard scope law.

At the *mean* level, top- k is worst for $\alpha \in \{0.25, 0.50\}$, but at $\alpha=0.25$ the seed-wise order holds only 1/2 (one seed breaks the full ranking). At $\alpha=0.75$, means reverse for top vs. random (0.94 vs. 0.77) and order holds on 0/2 seeds, while bottom remains highest on average. We treat these as **illustrative** hints that high real mix can blunt the top- k penalty—not as a definitive scope condition.

4.5 Transfer checks (directional; $n=3$)

These are **not** powered significance tests; we report $\text{mean} \pm \text{std}$ and seed-wise order counts.

- **8D GMM** (`w2_fs_hd`): retention 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 (bottom/rand/top); order holds 3/3.
- **Digits-PCA** (`w2_fs_digits_pca`): 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order holds 3/3.

An earlier Digits CVAE+classifier probe is inconclusive and unused for claims.

Table 4: Sliced W_2 contrast top–rand (positive = top worse). Bootstrap-style reporting: mean, std, count of positive cells, Wilcoxon p .

Gen. g	Mean ΔW_2	Std	$\#\{> 0\}$	Wilcoxon p
1	+0.113	0.099	8/8	0.014
2	+0.061	0.060	6/8	0.042
3	+0.039	0.035	7/8	0.030
4	+0.027	0.032	7/8	0.030
5	+0.024	0.033	7/8	0.030

Table 5: Mean retention r by real-mix α (two seeds). “Order holds” counts seeds with bottom>rand>top; the order is *not* guaranteed on both seeds except at $\alpha=0.5$.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	2/2
0.75	1.34	0.77	0.94	0/2

4.6 Selection-pool size (k_{frac})

Under $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$, $k_{\text{frac}} \in \{0.25, 0.5\}$ are operationally equivalent when n_{syn} binds; at 0.75 top- k remains worst.

5 Analysis

Likelihood top- k preferentially retains points the model already explains well (majority modes), reducing minority inflow. Bottom- k does the opposite relative to random. Aggregate sliced W_2 is a poor monitor for minority survival in our setting. The $\rho=5$ vs. $\rho=10$ split suggests the ordering is clearest when minorities remain above a near-floor mass at $g=0$.

6 Limitations

- Toy 2D/8D GMMs and Digits-PCA; not ImageNet-scale.
- Primary Wilcoxon $n=8$; Holm-adjusted $p \approx 0.085$. The seed-wise full order is robust at $\rho=5$ only.
- α ablation uses only two seeds (illustrative).
- Transfer checks: three seeds, no formal tests.
- For **double-blind** venues, the public GitHub URL reveals author identity; use an anonymous mirror for review (arXiv submission may keep the named repo).
- Contribution is a controlled empirical contrast relative to MAD/LSF/Feng, not a SOTA training method.

7 Conclusion

Under matched budgets, unlabeled likelihood ranking induces a minority-retention order that is **stable at moderate imbalance** ($\rho=5$) and **not seed-wise reliable at** $\rho=10$ in our matrix. Top- k does not improve sliced W_2 vs. random- k on the primary eight cells. Preferring “most likely” synthetics is not a free lunch for minority survival when rare modes are still measurable.

References

- Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. Self-consuming generative models go MAD, 2023.
- Zhongteng Cai, Yaxuan Wang, Yang Liu, and Xueru Zhang. Stabilizing self-consuming diffusion models with latent space filtering, 2025.
- Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. Beyond model collapse: Scaling up with synthesized data requires verification, 2024.
- Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Dhruv Pai, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David Donoho, and Sanmi Koyejo. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data, 2024.
- Peizhuo Li, Emre Aksan, Alexandru-Eugen Ichim, Thabo Beeler, and Olga Sorkine-Hornung. Diversify diffusion with temperature sampling and variance-corrective time shifting, 2026.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget, 2023.
- Yanbo Xu, Yu Wu, Sungjae Park, Zhizhou Zhou, and Shubham Tulsiani. Temporal score rescaling for temperature sampling in diffusion and flow models, 2025.

A Reproducibility and code

Cross-platform entry point: `python scripts/reproduce_w2.py` (also `scripts/reproduce_w2.sh` / `scripts/reproduce_w2.ps1`). All tables come from logged JSONL via analysis scripts. Code: <https://github.com/muhammadalahzeb/likelihood-rank-self-consuming-diffusion>.